

DATA 201 Group Project

SQL Analysis & Analytical Queries

Question #1 (Basic): What is the total listing count and average price across each room type?

SQL Query

```
USE group_project;
SELECT
    room_type,
    COUNT(*) AS total_listings,
    ROUND(AVG(price), 2) AS avg_price
FROM Listing
GROUP BY room_type
ORDER BY total_listings DESC;
```

Result

```
1 • USE group_project;
2 • SELECT
3     room_type,
4     COUNT(*) AS total_listings,
5     ROUND(AVG(price), 2) AS avg_price
6 FROM Listing
7 GROUP BY room_type
8 ORDER BY total_listings DESC;
```

100% 30:8

room_type	total_listings	avg_price
Entire home/apt	48006	476.76
Private room	15369	182.91
Hotel room	879	591.24
Shared room	373	74.91

Result Grid
Form Editor

Analysis & Key Takeaway

Entire homes/apartments constitute roughly three out of every four listings in the tri-city market, commanding average rates more than 2.6 times higher than private rooms (\$476.76 vs. \$182.91). Hotel rooms represent a small inventory fraction (879 listings) but carry the highest mean nightly rate (\$591.24).

Question #2 (Basic): How many listings in each city require a minimum stay of 30 nights or more?

SQL Query

```
USE group_project;
SELECT
    c.city_name,
    COUNT(l.listing_ID) AS long_term_listings
FROM Listing l
JOIN Neighbourhood n ON l.neighbourhood_id = n.neighbourhood_id
JOIN City c ON n.city_ID = c.city_ID
WHERE l.minimum_nights >= 30
GROUP BY c.city_name
ORDER BY long_term_listings DESC;
```

Result

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```
USE group_project;
SELECT
    c.city_name,
    COUNT(l.listing_ID) AS long_term_listings
FROM Listing l
JOIN Neighbourhood n ON l.neighbourhood_id = n.neighbourhood_id
JOIN City c ON n.city_ID = c.city_ID
WHERE l.minimum_nights >= 30
GROUP BY c.city_name
ORDER BY long_term_listings DESC;
```

100% 19:1

Result Grid Filter Rows: Search Export

city_name	long_term_listin...
Los Angeles	20481
San Diego	3861
San Francisco	2951

Result Grid Form Editor

Analysis & Key Takeaway

Los Angeles holds an extensive portfolio of long-term listings (20,481 units, representing ~46.6% of its market). This substantial concentration reflects hosts adopting minimum 30-night thresholds to classify their units under traditional tenant law rather than city short-term rental permit guidelines.

Question #3 (Advanced): Which hosts manage listings across more than one city, and how many total properties do they manage?

SQL Query

```
USE group_project;
SELECT
    h.host_name,
    COUNT(DISTINCT n.city_ID) AS num_cities,
    COUNT(l.listing_ID) AS total_listings
FROM Host h
JOIN Listing l ON h.host_ID = l.host_ID
JOIN Neighbourhood n ON l.neighbourhood_id = n.neighbourhood_id
GROUP BY h.host_ID, h.host_name
HAVING COUNT(DISTINCT n.city_ID) > 1
ORDER BY total_listings DESC
LIMIT 5;
```

Result

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```
USE group_project;
SELECT
    h.host_name,
    COUNT(DISTINCT n.city_ID) AS num_cities,
    COUNT(l.listing_ID) AS total_listings
FROM Host h
JOIN Listing l ON h.host_ID = l.host_ID
JOIN Neighbourhood n ON l.neighbourhood_id = n.neighbourhood_id
GROUP BY h.host_ID, h.host_name
HAVING COUNT(DISTINCT n.city_ID) > 1
ORDER BY total_listings DESC
LIMIT 5;
```

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Result Grid

Filter Rows: Search

Export: Fetch rows:

host_name	num_cities	total_listings
Blueground	3	444
RoomPicks	3	265
Evolve	2	221
Luxury Bookings Fze	2	192
RoomPicks	3	165

Result Grid Form Editor

Analysis & Key Takeaway

Only 142 hosts out of 30,338 operate across city boundaries. Multi-market operators are dominated by institutional brands like Blueground, confirming that cross-city operations are almost entirely the domain of corporate leasing and property management firms.

Question #4 (Advanced): Which neighbourhoods with at least 100 listings have an average price higher than the overall average price of all listings?

SQL Query

```
USE group_project;
SELECT
    n.neighbourhood_name,
    c.city_name,
    COUNT(l.listing_ID) AS total_listings,
    ROUND(AVG(l.price), 2) AS avg_neighbourhood_price
FROM Listing l
JOIN Neighbourhood n ON l.neighbourhood_id = n.neighbourhood_id
JOIN City c ON n.city_ID = c.city_ID
GROUP BY n.neighbourhood_name, c.city_name
HAVING COUNT(l.listing_ID) >= 100
      AND AVG(l.price) > (SELECT AVG(price) FROM Listing)
ORDER BY avg_neighbourhood_price DESC
LIMIT 5;
```

Result

1 • USE group_project;

2 • SELECT

3 n.neighbourhood_name,

4 c.city_name,

5 COUNT(l.listing_ID) AS total_listings,

6 ROUND(AVG(l.price), 2) AS avg_neighbourhood_price

7 FROM Listing l

8 JOIN Neighbourhood n ON l.neighbourhood_id = n.neighbourhood_id

9 JOIN City c ON n.city_ID = c.city_ID

10 GROUP BY n.neighbourhood_name, c.city_name

11 HAVING COUNT(l.listing_ID) >= 100

12 AND AVG(l.price) > (SELECT AVG(price) FROM Listing)

13 ORDER BY avg_neighbourhood_price DESC

14 LIMIT 5;

75% 9:14

Result Grid

Filter Rows: Search

Export: Fetch rows:

neighbourhood_name	city_name	total_listings	avg_neighbourhood_pr...
Beverly Crest	Los Angeles	216	1878.92
Malibu	Los Angeles	410	1823.61
Hollywood Hills West	Los Angeles	668	1425.16
Financial District	San Francisco	300	1173.08
Unincorporated Santa Monica Mountains	Los Angeles	199	1020.96

Result Grid

Form Editor

Analysis & Key Takeaway

Integrating a scalar subquery (SELECT AVG(price) FROM Listing) within the HAVING clause establishes a dynamic baseline without hardcoding cutoff values. The threshold filter (COUNT >= 100) removes low-sample outliers and pinpoints high-density luxury enclaves in Los Angeles and the business core of San Francisco that sustain average rates four to five times higher than typical regional listings.